EXHIBIT 2

IN THE UNITED STATES DISTRICT COURT
FOR THE EASTERN DISTRICT OF LOUISIANA

KATHLEEN MOLLER,	:	
	:	
Plaintiff,	:	
	:	Civil Action No.
vs.	:	
	:	2:24-cv-228-WBV-DPC
MARTIAN SALES, INC.,	:	
et al.,	:	
	:	
Defendants.	:	

VIRTUAL VIDEOTAPED DEPOSITION OF
JOSEF THUNDIYIL, MD, MPH
Friday, January 16, 2026
10:08 a.m. Eastern Standard Time

REPORTER: Dawn A. Jaques, CSR, CLR

DIGITAL EVIDENCE GROUP
1730 M Street, NW, Suite 812
Washington, D.C. 20036
(202) 232-0646

kratom as a dirty drug?

A You could ask him if he used that term, yes. I don't know if he would recall it.

Q What amount of kratom causes toxicity?

A Well, I think it depends on a lot of factors. We don't know the exact level that causes toxicity. We have some different estimates.

For example, you know, we know when we look at what's known as the IC50 for the potassium current, so the concentration at which about 50% of people will experience some blockade of the inward cardiac potassium channel, that number is estimated to be about 0.3 mol/L, which is a concentration of about 17 ng/mL.

So there's a potential for harm at levels -- or at least for 50% of people to experience that. We have seen deaths at lower concentrations.

Because mitragynine is not standardized -- or kratom is not standardized in its formulation, it's nearly impossible at this point, with the existing science, to have a dose that we know causes harm. Most of what we try to do is extrapolate from different effects.

The other complicating factor is the fact that different formulations seem to have effects. It could be dried leaf, it could be extract. These often get mixed up or confused, and there doesn't seem to be a standardization. Unlike a lot of drugs, which might be one chemical, kratom has, as we've said before, about 40 different alkaloids.

So all these make it difficult to have a specific dose or concentration that we know will absolutely cause harm, but we know that inter-individually from different drugs, like other opiates, that some people experience harm at a low level; some people, due to tolerance and other factors, won't

BY MS. PAYTON:

Q And is it your opinion that Ms. Moller experienced respiratory depression as a result of kratom?

MR. COWGILL: Objection to form.

THE WITNESS: Ultimately she died, and what we know is that her heart stopped.

What I can't tell for certain is whether her heart stopped and then she stopped breathing, or she stopped breathing and then her heart stopped.

There is mechanistic support for both, and I don't know -- there's no way to really know for sure which one caused her death.

BY MS. PAYTON:

Q You also opine that kratom can lead to cardiac arrhythmia, correct?

A Yes.

Q Again, are we able to say at what level?

A The only piece that we can say, we know that there's a dose dependent increase in QT prolongation. This has been shown in one of the Singh-Adbullah studies.

We know that, from what I mentioned earlier, that at concentration about 127 ng/mL, there is the potential about 50% that's -- 50% the max concentration, inhibitory concentration, or the inward potassium current.

We don't know the translation. It's not possible yet to back extrapolate what that dose might be because it's not a listed thing on the products of mitragynine, but we know with that concentration and with escalating doses, and especially in people who might be taking therapeutic doses of other drugs, like Phenergan and Benadryl in her case, that that would significantly aggravate her risk, putting her at high risk. To have two or three, now three QT prolonging drugs, is what would put someone at risk for torsades, or

fatal arrhythmia.

Q What other causes of cardiac arrhythmia do you commonly see in your practice?

A Every cause. Cardiac arrest, ischemia, medication induced, overdose, congenital heart.

Yeah, I've seen probably nearly every common cause of cardiac arrest and been part of that resuscitation.

Q So lots of things can cause a cardiac arrhythmia?

A Yeah. And for that reason, I included all these in my differential diagnosis when considering that.

Q And you considered Ms. Moller's particular heart that she had, right?

A Absolutely, yeah.

Q And abnormalities in that heart?

A Yeah. I looked at what was described as this -- I guess it was a high takeoff with aortic flap, and I looked at the

MR. COWGILL: Objection, form.

THE WITNESS: Well, my interpretation of the evidence so far is that it doesn't look like it does, but I will say with the caveat that so little is still known about kratom and mitragynine and its 40 different alkaloids that are in there that the true scientific answer is still unknown.

BY MS. PAYTON:

Q If, hypothetically, the true scientific answer was there is postmortem redistribution, then Ms. Moller's premortem concentration would have been significantly lower than 150 or so nanograms per milliliter, correct?

MR. COWGILL: Objection to form, calls for speculation.

THE WITNESS: No, it can actually go both ways. Sometimes it could have been quite a bit higher. So it's not a unidirectional process.

Q Okay. And you agree with me the Huestis study is totally separate from this FDA study, right?

A Right, sure, it's separate. It has some similarities, but yes.

Q But different subjects, different dataset, everything?

A Yeah.

Q Okay. Do you intend to offer any opinions in this case other than the ones in your report?

A Well, yeah, I think all the ones we discussed.

I think we didn't really talk a lot about the process of my -- the process by which I developed my opinions, and causation, I don't think we got into that, but those would be things that, if asked, I would talk about.

Q What was the process by which you -- oh, sorry, I cut you off.

A No, okay. If you want to go ahead

first, then I can talk.

And the other things would be any disagreements I had with some of the defense experts' reports, some of which we talked about, but I don't think we got through them all.

Q Do you -- what was the process by which you came to your opinion in this matter?

A So we touched on this earlier a little bit, but it was really a two-step process.

The first is determining -- and I think this is, in my mind, well established -- whether or not it's even possible for kratom to cause death.

And in order to do that, I relied on a well recognized causation process known as the Bradford Hill criteria, which includes steps such as identifying plausibility, specificity, dose response, analogy, coherence with existing literature, experimental evidence, strength of association, as well as

temporality and consistency.

So those -- I took it through each of those nine criteria to establish that there is a general -- there is general causation supported.

Now, the second part of my process was then applying it to Harmony Moller, and in order to do that, I used a well recognized process in medicine where we are attempting to come to a diagnosis, sometimes just referred to as the differential diagnosis.

In applying it to toxicology, first I had to decide whether or not there was an exposure, identifying the temporal relationship. Did the exposure happen before the outcome? Was the exposure high enough to cause it? Is it consistent with our understanding of the science? And, finally, what are the other potential causes? And that's where the differential diagnosis comes in.

I think we covered most of the

discussed in the discussion sections of several papers, including the plausibility mechanism, the dose response, but I haven't seen someone take it through it step by step. I don't know the reason.

Q Did you do that step-by-step analysis in writing?

A No.

Q So just something you did mentally in your head?

A Yes. And I do this with just about any substance that has the potential to cause harm if I'm asked to determine causation, or if internally I want to decide if there is a possibility of a causal relationship.

Q Did you do that analysis for this case, or had you already done it with earlier kratom cases?

A Yeah, I had done it for an earlier kratom case. I think since that case, additional literature, including my own, increased experienced, and my own study with